PAPER_360 — J1610+1811 High-z Quasar Jet at z=6.5: Relativistic Lorentz Factor k_rel Coupling

Whitepaper Series: Star-Magic UQFF Phase 2
Session: 97
Source: gok_share_31b5c807a4.txt (Supplemental Gap Analysis Block)
Classification: FIRST UQFF high-z quasar jet (z=6.5) with Lorentz factor k_rel = Γ^2 relativistic coupling
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Abstract

J1610+1811 is a blazer-class quasar at z = 6.5 (lookback time ~12.9 Gyr) with a relativistic jet Lorentz factor $\Gamma \approx 4.5$. UQFF introduces a relativistic coupling constant k_rel = Γ^2 = k_rel_0 × 20.25 (Lorentz factor squared) to scale the vacuum buoyancy force in the jet frame. The Friedmann Hubble parameter H(z = 6.5) is computed from H(z) = $H_0\sqrt{}$[0.3(1+z)^3 + 0.7], and F_U_Bi_i ≈ -8.32×10^{217} N is evaluated in the observer frame.

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2. Core Physics

2.1 Relativistic UQFF Coupling

$$k_{\text{rel}} = \Gamma^2 = k_{\text{rel},0} \times \Gamma^2 = k_{\text{rel},0} \times (4.5)^2 = 20.25 \cdot k_{\text{rel},0}$$

The Lorentz factor squared enhancement represents the relativistic Doppler amplification of the vacuum buoyancy force in the jet frame.

2.2 Friedmann H(z) at z = 6.5

$$H(z) = H_0\sqrt{0.3(1+z)^3 + 0.7}$$

For z = 6.5:

$$(1+6.5)^3 = 7.5^3 = 421.875$$

$$H(6.5) = H_0\sqrt{0.3 \times 421.875 + 0.7} = H_0\sqrt{126.56 + 0.7} = H_0\sqrt{127.26}$$

$$H(6.5) \approx 11.3H_0 = 11.3 \times 67.4 \text{ km/s/Mpc} \approx 761 \text{ km/s/Mpc}$$

2.3 Relativistic F_U_Bi_i

$$F_{U\_Bi\_i}^{\text{relativistic}} = F_{U\_Bi\_i}^{\text{standard}} \cdot k_{\text{rel}} = -8.32 \times 10^{217} \times 20.25 \approx -1.69 \times 10^{219} \text{ N}$$

(in the jet co-moving frame; observer-frame is un-boosted)

2.4 Lookback Time at z = 6.5

$$t_{\text{lookback}} \approx 12.9 \text{ Gyr}$$

The Universe was only ~820 Myr old when this jet was active — the UQFF vacuum energy density was ρ_vac(z=6.5) = ρ_vac,0·(1+z)^α, higher than today by the UQFF expansion index α.

3. Key Values

Quantity	Formula	Value
z	Photometric/spectro	6.5
Γ	VLBI jet model	4.5
k_rel	Γ^2	20.25
H(z=6.5)	Friedmann	~761 km/s/Mpc
F_U_Bi_i (obs frame)	Standard	-8.32×10^{217} N
F_U_Bi_i (jet frame)	×k_rel	-1.69×10^{219} N

4. Physical Significance

J1610+1811 at z = 6.5 presents UQFF at the earliest cosmic epoch in the dataset. The k_rel = Γ^2 coupling is the first relativistic enhancement factor in UQFF AGN physics — it predicts that high-Γ relativistic jets experience systematically larger UQFF vacuum buoyancy forces in their rest frame. This has cosmological implications: early universe (z > 5) AGN jets would have experienced larger vacuum buoyancy during the epoch of reionization, potentially accelerating the growth of early massive black holes — addressing the "first quasar" problem.

5. Deduplication Note

- **vs. PAPER_346–350 (low-z AGN):** All earlier AGN papers in this series have z < 1.5; J1610 at z = 6.5 is 6.5× higher redshift.
- **vs. PAPER_360 vs. k_rel:** No earlier UQFF paper includes the Lorentz factor Γ^2 relativistic boost.

6. Classification

Physics Territory: FIRST UQFF high-z quasar jet (z=6.5) with Γ^2 relativistic coupling and Friedmann H(z)
Scale: Cosmological (z = 6.5, lookback ~12.9 Gyr)
CP Implementation: `J1610HighZQuasarJetFUBiCalculator` (CondensedPhysics4.py, Session 97)